

4 (a) State the first law of thermodynamics.

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..... [2]

(b) A sample of an ideal gas contains 3.58×10^{24} molecules. The sample is at a temperature of 298 K and a pressure of 1.84×10^5 Pa.

(i) Show that the volume of the gas is 0.0800 m^3 .

[2]

(ii) Calculate the amount of gas n in the sample.

$n = \dots\dots\dots \text{ mol}$ [1]

(iii) Determine the average translational kinetic energy E_K of one molecule of the gas.

$E_K = \dots\dots\dots \text{ J}$ [2]

(iv) Explain why the average potential energy of one molecule of the gas is zero.

.....
..... [1]

(v) Calculate the internal energy U_A of the gas.

$U_A = \dots\dots\dots \text{ J}$ [1]

(c) The gas in 4(b) undergoes two changes AB and BC, as shown in Figure 4.1.

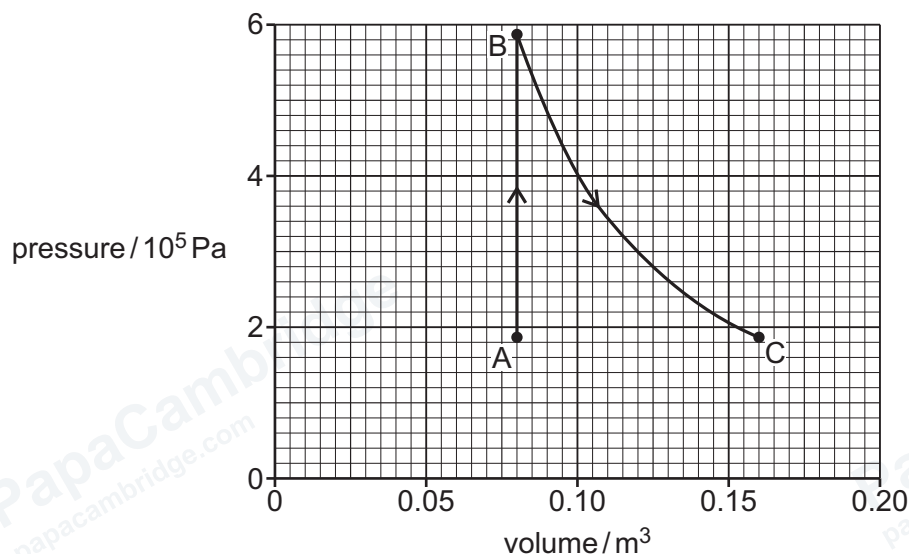


Figure 4.1

In change AB, the gas is heated at constant volume from its initial pressure of $1.84 \times 10^5 \text{ Pa}$ to a pressure of $5.84 \times 10^5 \text{ Pa}$.

In change BC, the gas rapidly expands and returns to its original pressure with no transfer of thermal energy. The volume of the gas after this expansion is 0.160 m^3 .

- (i) Determine an expression, in terms of U_A , for the thermal energy Q supplied to the gas during change AB. Explain your reasoning.

$Q = \dots\dots\dots$ [2]

- (ii) Determine an expression, in terms of U_A , for the work W done on the gas during change BC. Explain your reasoning.

$W = \dots\dots\dots$ [2]

[Total: 13]